WEEK-4 ASSIGNMENT

Title: Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

Assignment Tasks

Task 1:

- Explore Azure Portal
- Create a Resource Group

Deliverable:

✓ **Explore Azure Portal**

Microsoft Azure is a comprehensive cloud computing platform by Microsoft. It allows individuals and businesses to build, deploy, and manage applications and services over the internet, eliminating the need to own and maintain physical servers. Users pay only for the computing resources they actually consume.

Core Components :

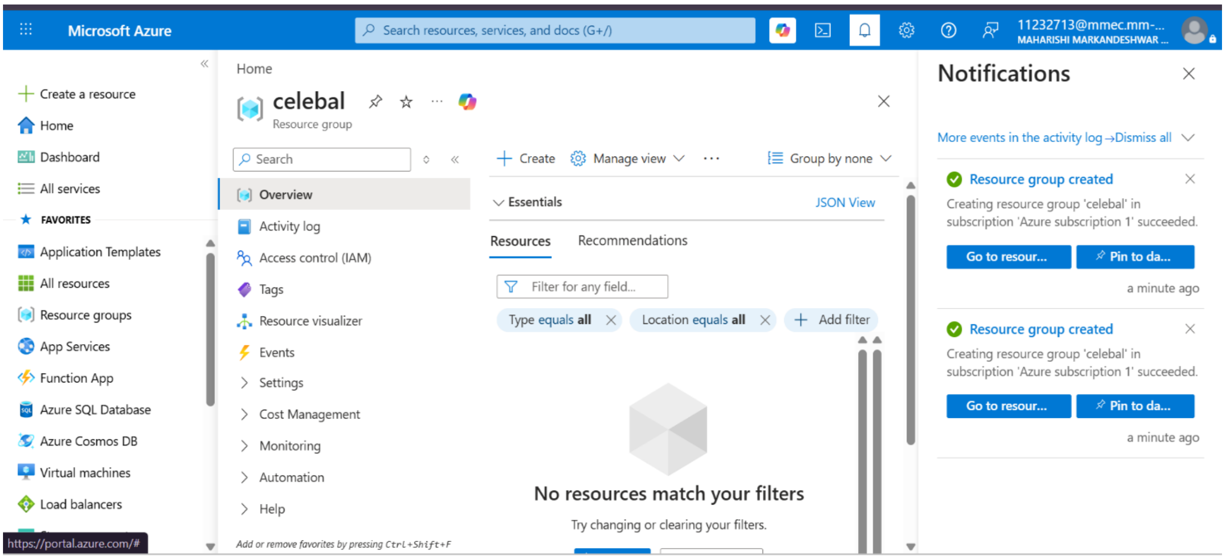
- **Top Search Bar:** Located at the top center, used for quickly finding resources, services, documentation, and deployment guides.
- **Navigation Panel:** Located on the far left, offering quick access to "Home", "Dashboard", and a list of bookmarked "Favorites" like App Services, Virtual Machines, and Resource Groups.
- **Notification Pane:** Located on the far right, displaying real-time status updates, success logs, and background activity trackers for ongoing deployments.

✓ **Create a Resource Group**

A Resource Group is a logical container or folder deployed within an Azure subscription. It aggregates related cloud resources—such as virtual machines, databases, virtual networks, and storage accounts—that share a common lifecycle, allowing you to manage, secure, and monitor them as a single entity.

Implementation:

- ○ Log into the Microsoft Azure Portal.
- ○ Select **Resource groups** from the left navigation menu or search bar.
- ○ Click the **+ Create** button at the top-left.
- ○ Select your **Subscription** (e.g., *Azure subscription 1*).
- ○ Enter a unique **Resource Group Name** (e.g., *celebal*).
- ○ Choose your deployment **Region**.
- ○ Click **Review + create**, then click **Create** once validation passes.
- ○ Open the **Notifications** pane (bell icon) and click **Go to resource** on the success message.



Task 2: Storage Setup

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable:

✓ **Create a Storage Account**

An **Azure Storage Account** is a globally unique management container that groups all of your Azure Storage data services together. It provides a secure, massively scalable, and highly available namespace for accessing your data over HTTP/HTTPS from anywhere in the world.

Implementation:

- ○ In the Azure Portal search bar, type "Storage accounts" and select it.
- ○ Click the **+ Create** button.

- Select your Subscription and Resource Group.
- Enter a globally unique name (in my case named as blobcontainercelebal).
- Choose your deployment region and select either *Standard* or *Premium* performance.
- Click **Review + create**, then click **Create** once validation passes.

✓ **Create a Blob Container**

A Blob Container acts like a directory or folder within an Azure Storage Account. It organizes a set of blobs (binary large objects) and enforces security and access policies on those specific files. All blobs must reside inside a container.

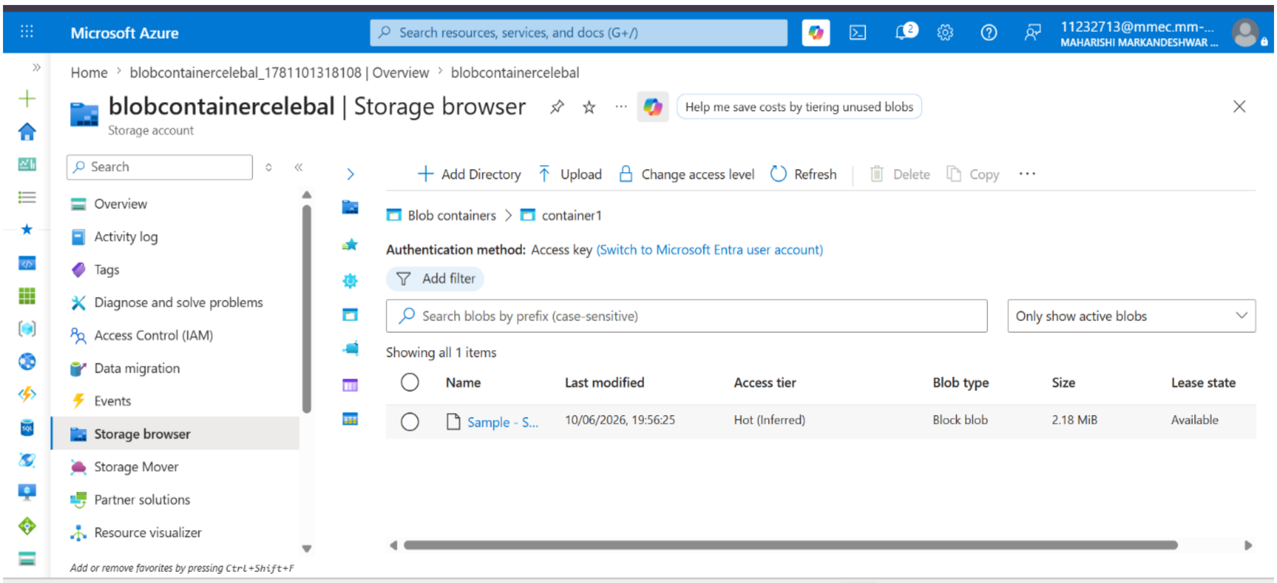
Implementation:

- Go to your newly created storage account (blobcontainercelebal).
- Click on **Storage browser** from the left-hand menu
- Click on the **Blob containers** directory item in the main view.
- Click the **+ Add container** button at the top.
- Name your container (your screenshot uses the name container1) and choose the public access level (Private, Blob, or Container). Click **Create**.

✓ **Upload a CSV File**

Implementation:

- Click on your container (container1) within the Storage browser.
- Click the **↑ Upload** button located on the top action toolbar.
- In the slide-out pane, browse your local computer and select your target CSV file.
- Click **Upload**.



Task 3: ADF Basics

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

Deliverable:

✓ **Create Azure Data Factory**

Azure Data Factory (ADF) is a cloud-based data integration service. It acts as an Extract-Transform-Load (ETL) and Extract-Load-Transform (ELT) platform. ADF allows you to orchestrate and automate data movement and data transformation at scale.

Implementation:

- Type "Data factories" into the global Azure Portal search bar.
- Click **+ Create** and assign your project details:
- Select your group (e.g., celebal).
- Give it a unique identifier (mycelebaldatafactoryadf).
- Select your closest region.
- After deployment finishes, open the resource and click **Launch Studio** to enter the dedicated ADF workspace.

✓ **Explore ADF UI (Author, Monitor, Manage)**

The ADF User Interface provides a specialized graphical workspace separate from the main Azure Portal. It organizes data engineering workflows into three distinct operational hubs :

- **Author (Pencil Icon):** The design canvas. Use this tab to visually build data pipelines, datasets, mapping data flows, and power queries.
- **Monitor (Gauge Icon):** The operation dashboard. Use this tab to track active or past pipeline runs, view execution logs, check trigger statuses, and analyze performance bottlenecks.

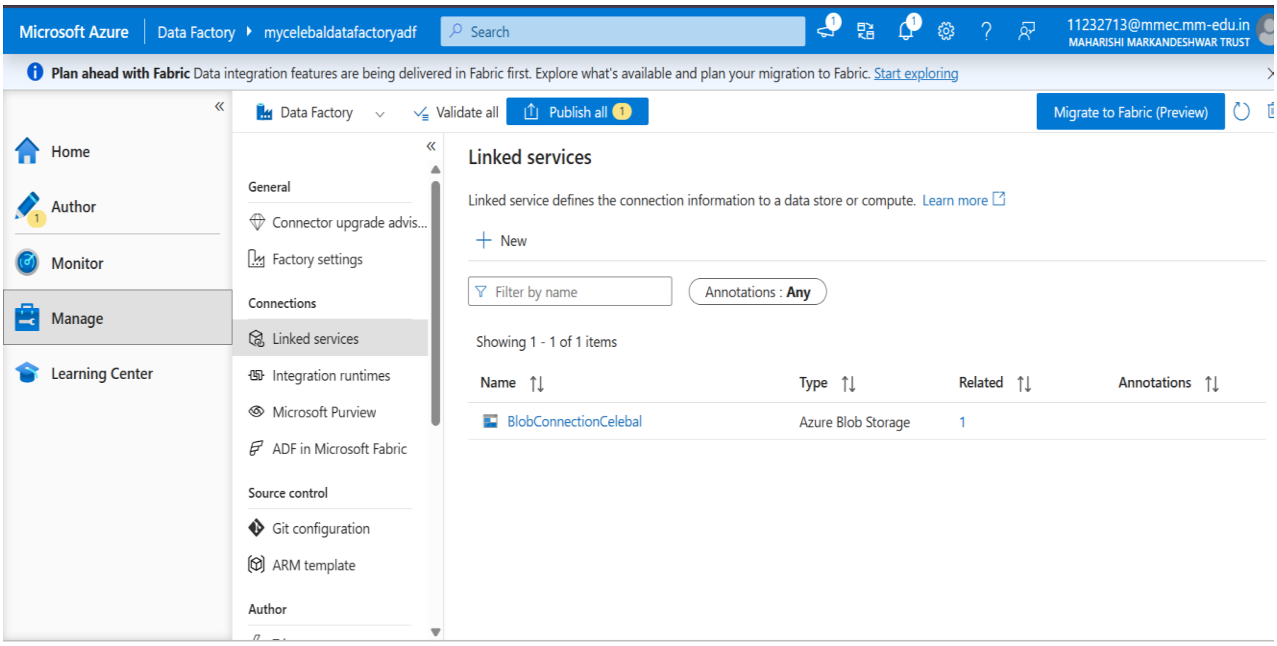
- **Manage (Toolbox Icon):** The management plane. As selected in your screenshot, this hub handles backend configurations including connections, security credentials, integration runtimes, and source control.

✓ **Create Linked Service (Blob Storage)**

A Linked Service is essentially a connection string within Azure Data Factory. It defines the explicit connection information and authentication mechanism required for ADF to securely connect to external data stores or compute resources.

Implementation:

- Click the **Manage** hub on the left menu, then select **Linked services** under the Connections submenu.
- Click the **+ New** button on the main toolbar.
- In the search drawer, choose **Azure Blob Storage** and click Continue.
- Name your connection (BlobConnectionCelebal).
- Select your **Integration Runtime** (default is AutoResolve).
- Choose your account selection method, pick your specific storage account (blobcontainercelebal), and test the connection.
- Click **Create**. Then, validate all.



✓ Create Azure Data Factory Datasets

An **Azure Data Factory Dataset** is a logical reference that identifies the specific structure and location of your data within a data store. It acts as a named view pointer for pipeline activities to know exactly where to read (source) or write (destination) files without embedding hardcoded paths into the processing code.

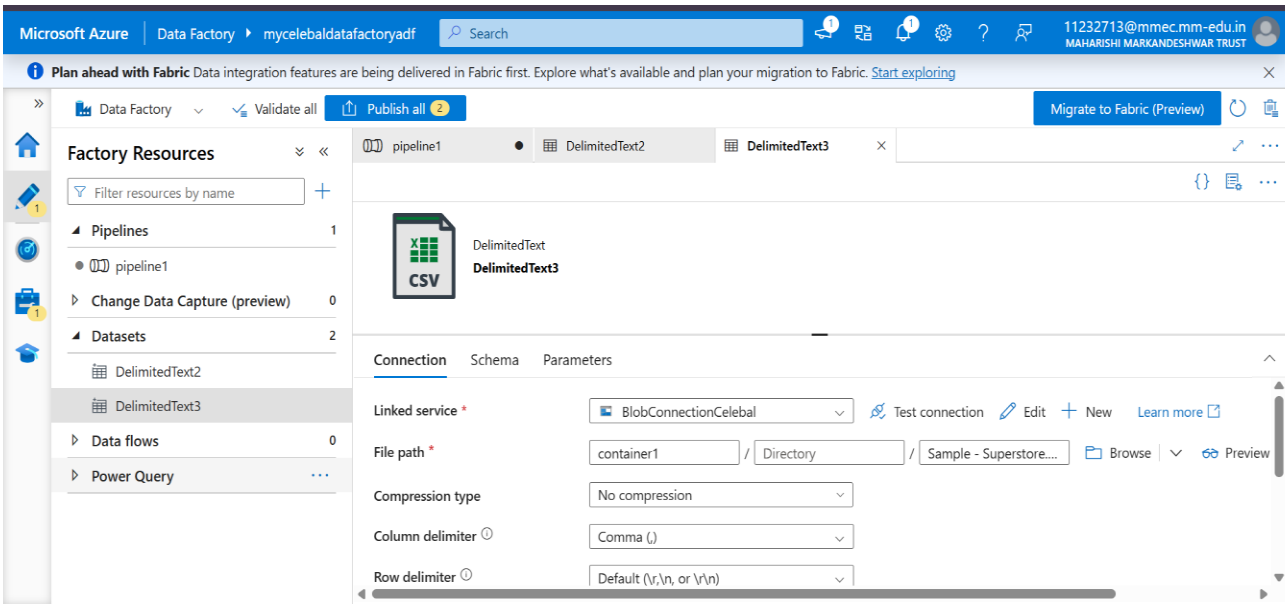
Implementation:

Source Dataset Configuration

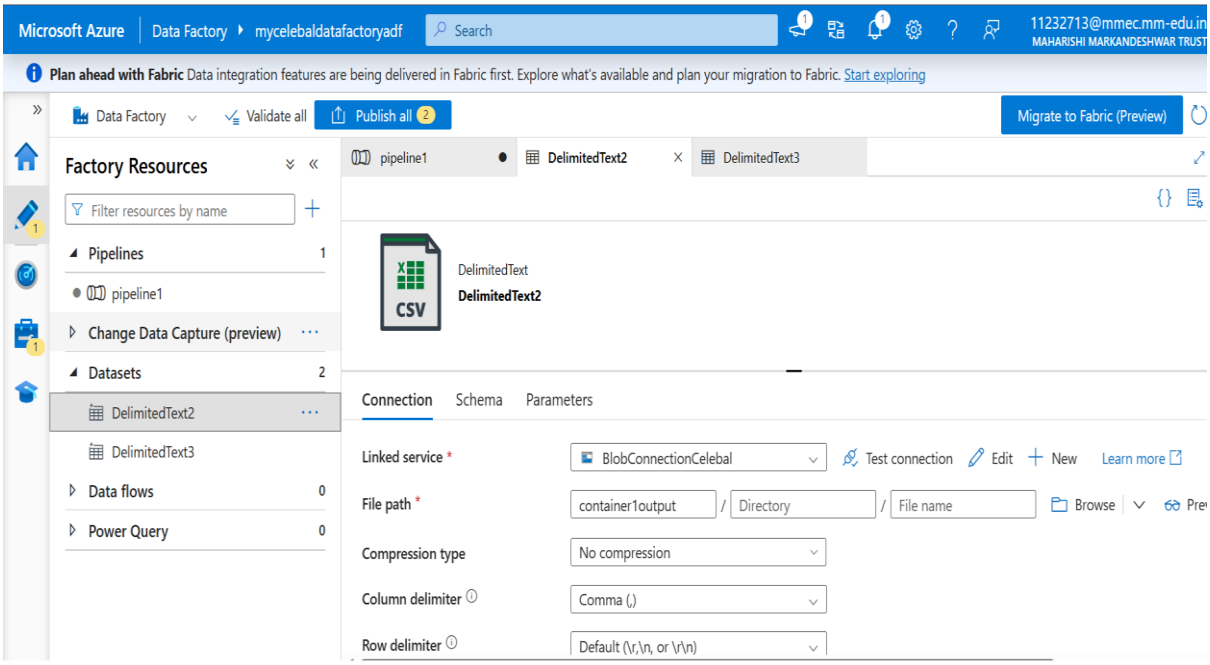
- Open Azure Data Factory Studio and click the **Author (Pencil)** icon on the far left.
- Expand the **Factory Resources** menu, click the **+** (plus) icon, and select **Dataset**.
- Choose **Azure Blob Storage** as your store type.
- Select **DelimitedText** (CSV) as the format type.
- Name your source dataset (as shown in your left pane, e.g., DelimitedText2).
- Select BlobConnectionCelebal from the **Linked service** dropdown menu.
- Under **File path**, use the browse tool to select your uploaded Sample - Superstore... file in the file box.
- Ensure the **Column delimiter** is explicitly set to **Comma (,)**.

Same for Destination Dataset

Source dataset



Destination dataset



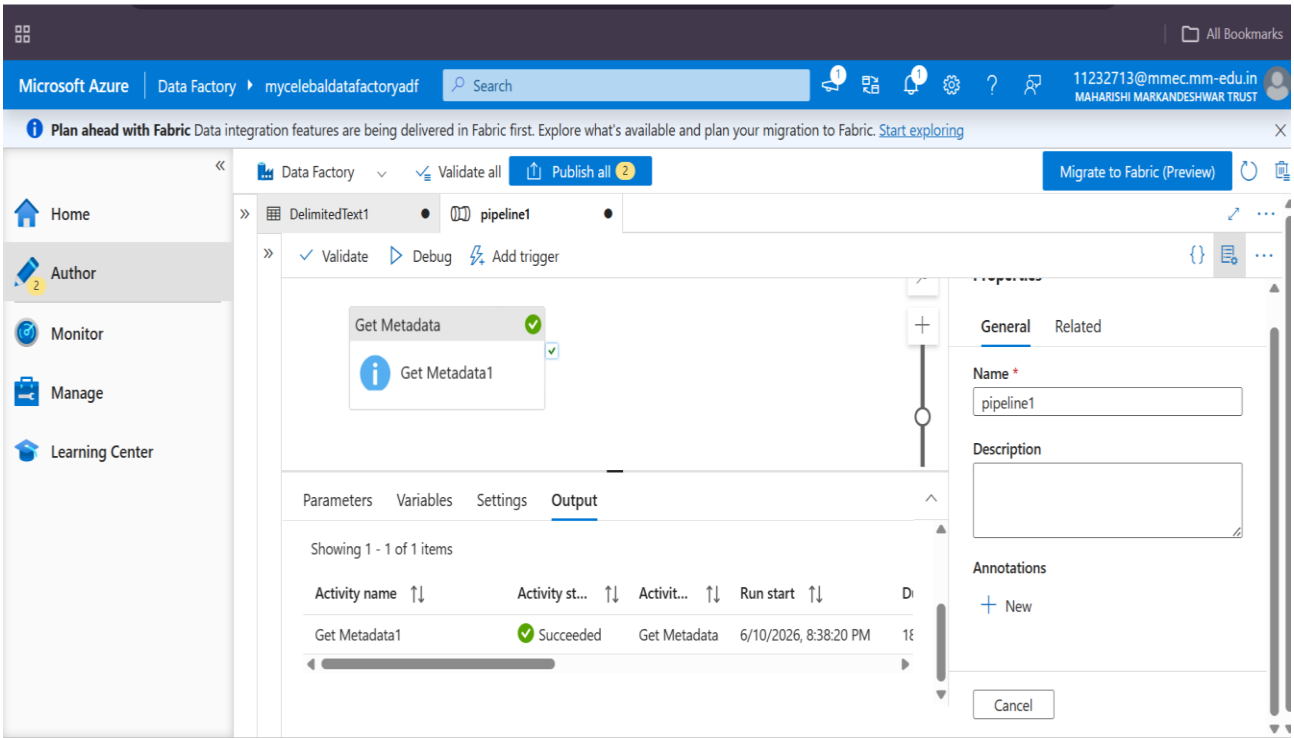
✓ **Use Get Metadata Activity**

The **Get Metadata activity** is a control flow activity in Azure Data Factory used to retrieve structured information about any target data source or file. It allows a pipeline to inspect file properties—such as existence, size, structure, child items, or modification dates—before executing subsequent transformation steps.

Implementation:

- Under the **Author** tab, open your existing pipeline (named pipeline1).
- Locate the **Activities** search bar on the canvas panel, type Get Metadata, and drag the activity onto the canvas surface.
- Select the activity to open its configuration pane.
- Under the **General** tab on the right side, change the name identifier to Get
- Switch to the **Settings** tab in the bottom configuration panel.
- Select your previously created source dataset (pointing to CSV in container1).
- Under **Field list**, click **+ New** to select Exists.
- Click the **Debug** button on the top toolbar of the pipeline canvas to execute a test run of the control flow.

- Verify the **Activity status** column shows a green checkmark icon with the status **Succeeded**.
- **Inspect Output Details**



Task 4: Pipeline Development

- Create pipeline using Copy Data activity
- Configure source and destination

Deliverable:

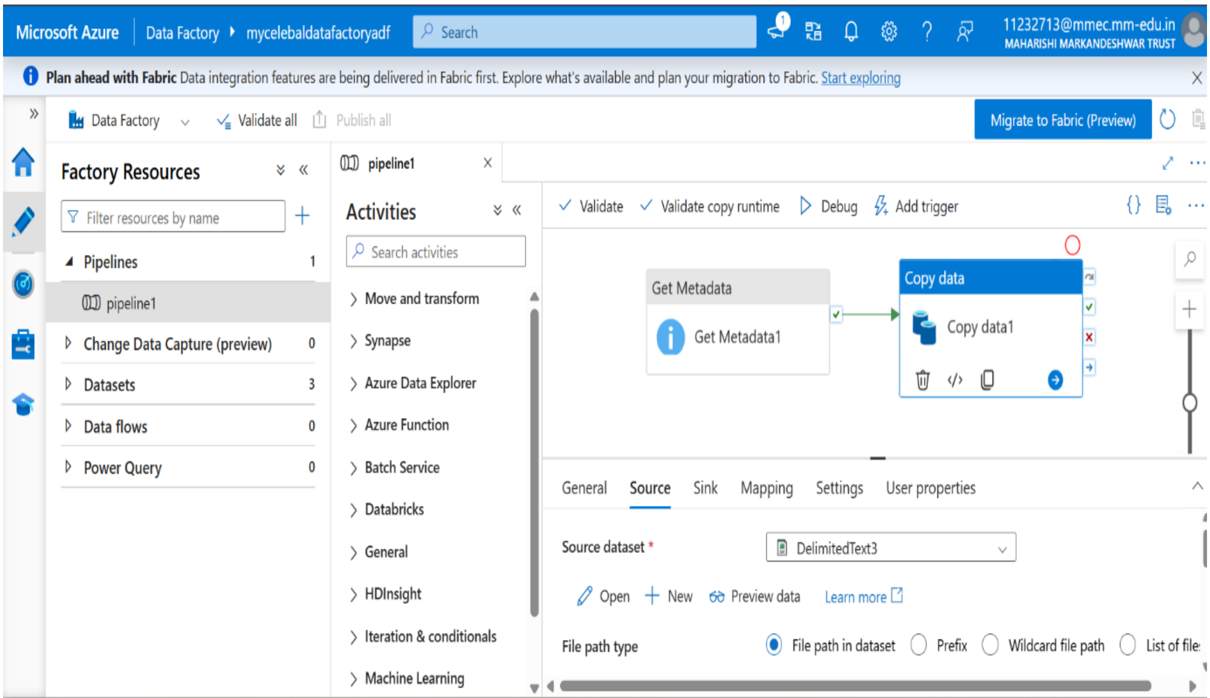
✓ **Copy Activity**

It moves data from a source data store to a destination (sink) data store, handling data serialization, compression, and column mapping seamlessly.

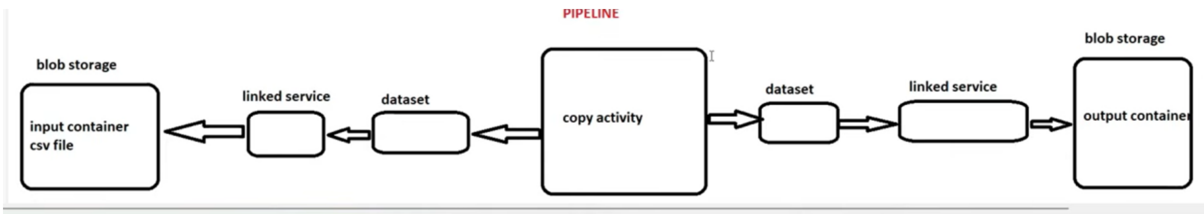
Implementation:

- Open pipeline1 from the Pipelines menu.
- Expand Move and transform in the activity panel and drag Copy data onto the canvas.
- Connect the green **Success arrow** from Get Metadata1 to Copy data1.
- Click Copy data1 → go to the Source tab → select your input dataset (DelimitedText3) from the dropdown.

- Destination: Switch to the Sink tab → select your output dataset from the dropdown.
- Click Validate All.



PIPELINE:



Task 5: Pipeline Execution

Do:

- Run pipeline using Debug/Trigger

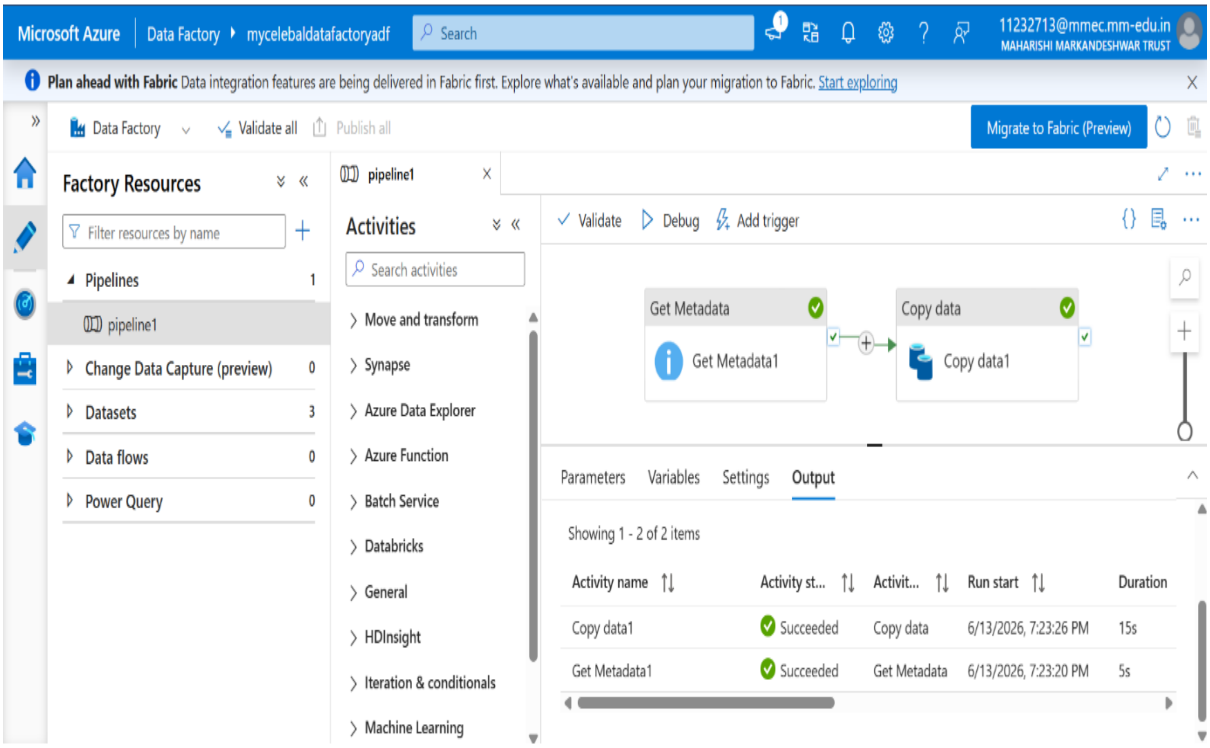
Deliverable:

- **Debug**: Runs a manual, real-time test of your pipeline directly from the canvas. It does not require you to save or publish changes first.
- **Trigger**: Executes the pipeline in production based on a configured automation (such as a time schedule, file arrival event, or window interval). It requires all changes to be published first.

Implementation:

- Click Debug on the pipeline toolbar to run an immediate test.

- Verify Results: Check the bottom Output tab to monitor the execution workflow.



Task 6: IAM Roles

Do:

- Assign roles:
  - Reader
  - Contributor
- Provide access to ADF for Storage

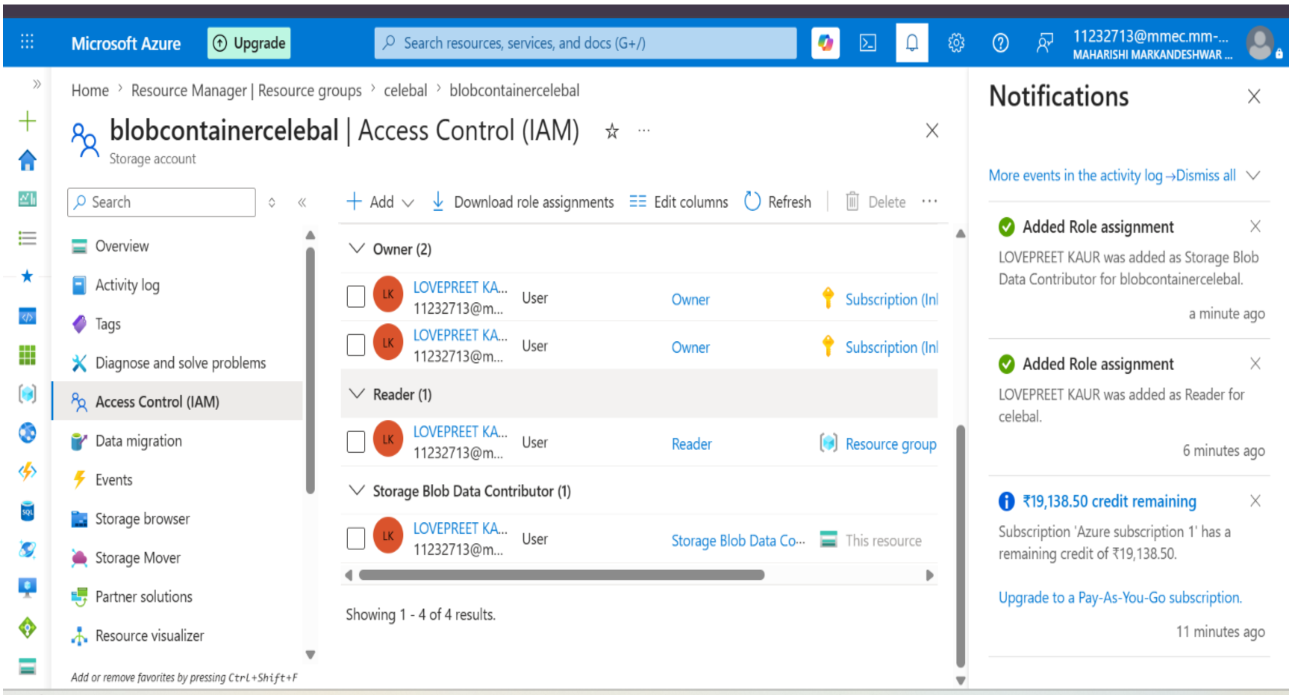
Deliverable:

- **Access Control (IAM)**: The portal interface used to manage permissions using Azure Role-Based Access Control (RBAC).
- **Reader**: Grants view-only access to resources without permission to modify them.
- **Storage Blob Data Contributor**: Grants full read, write, and delete permissions to blob container data.

Implementation

- Navigate to your storage account (blobcontainercelebal) and click Access Control (IAM) on the left menu.
- Click + Add → Add role assignment.

- Select Role: Choose Reader (next time same for Storage Blob Data Contributor), then click Next.
- Click + Select members, select the target identity (e.g., LOVEPREET KAUR), and click Review + assign.
- Check the Notifications pane to confirm both assignments succeeded.



Mini Project (Final Task)

Problem Statement:

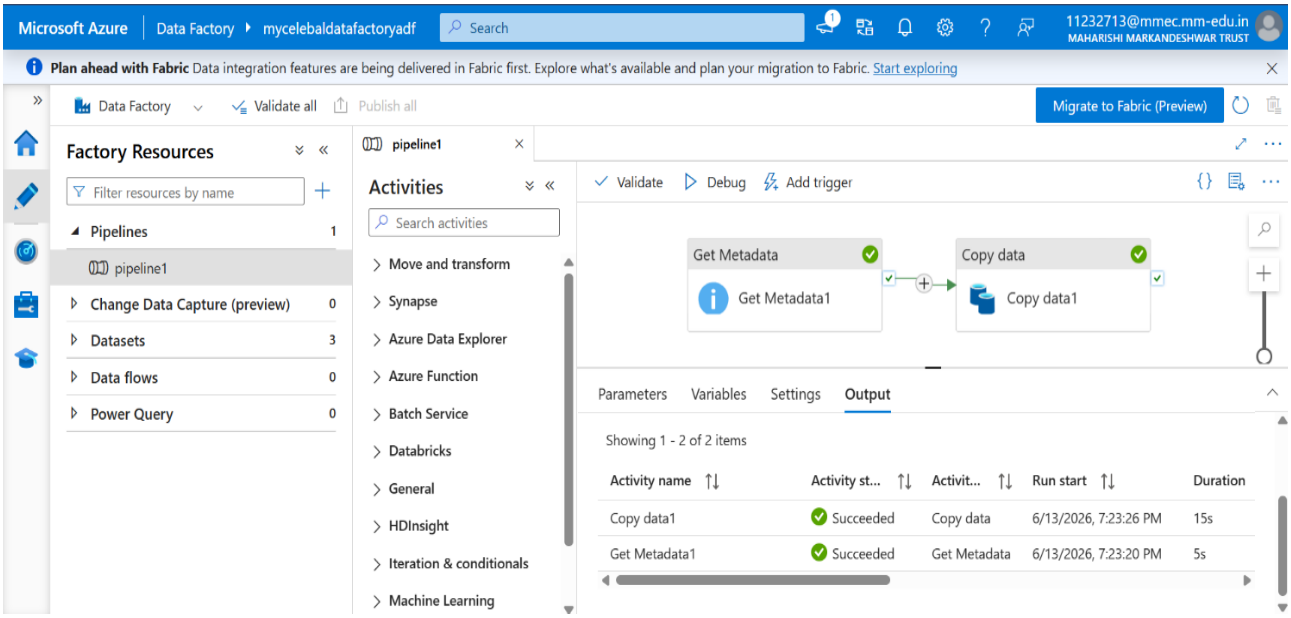
Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

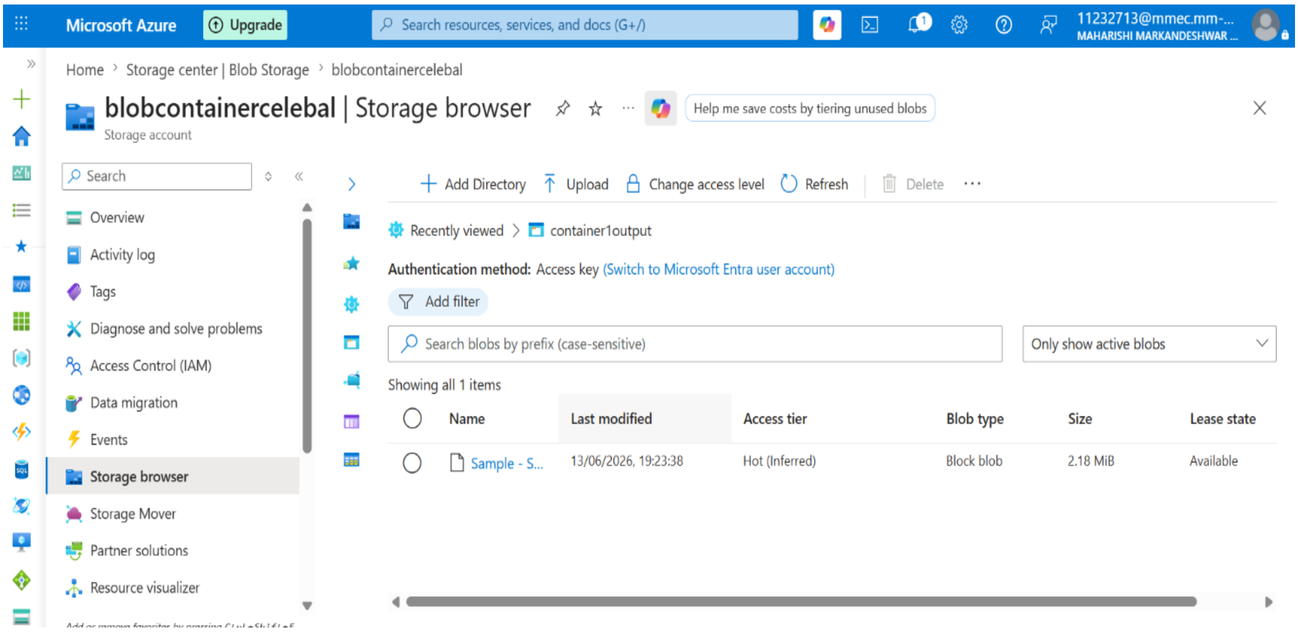
- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

- Pipeline executed successfully

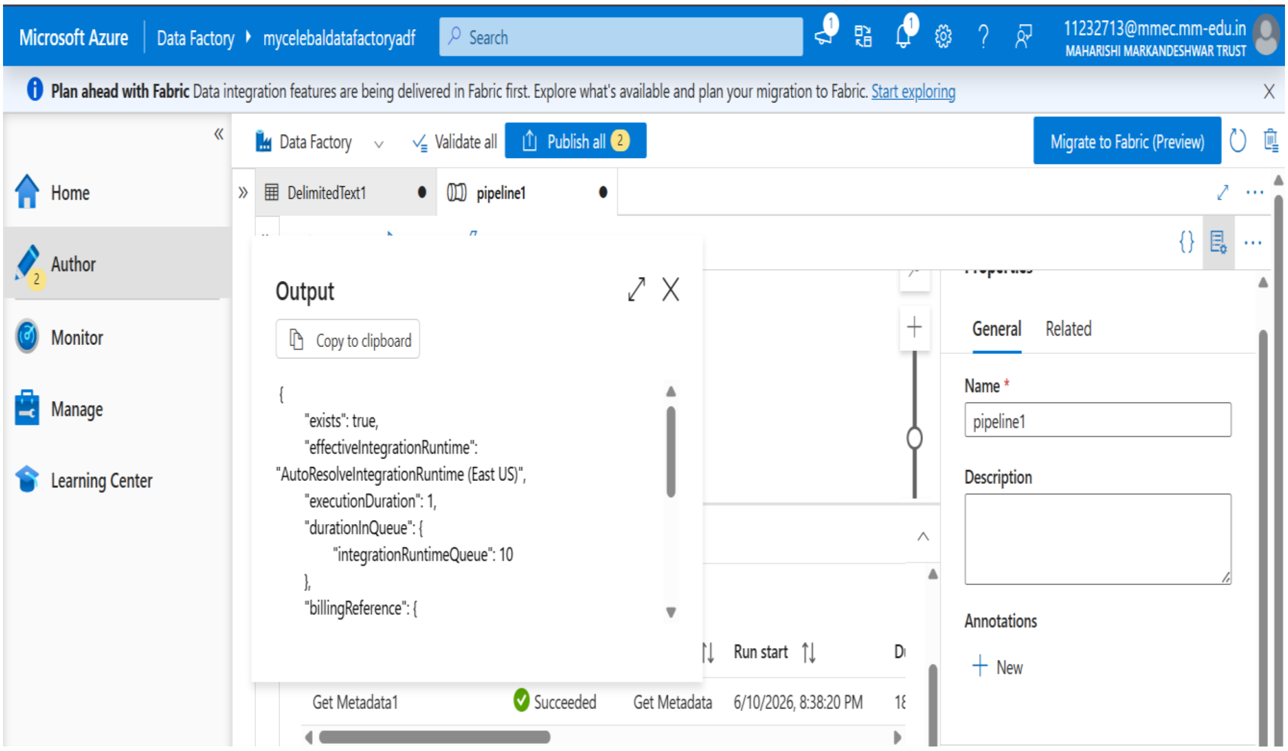


- Data copied to destination



Data copied to container1celebal blob container after pipeline execution.

- Metadata validated



Metadata validated : It checked that file exists.

❖ Conclusion:

- This assignment successfully demonstrates how to build and secure an end-to-end cloud data pipeline in Microsoft Azure. By connecting **Resource Groups**, **Blob Storage**, and **Azure Data Factory**, a complete data ingestion workflow was built.
- Using **RBAC role assignments** ensured secure access control, while ADF tools like **Linked Services**, **Datasets**, and **Copy Data activities** automated the file transfer process. Successfully running the pipeline via **Debug** confirms that the architecture is fully functional, secure, and ready for scalable enterprise data integration.